VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JnanaSangama", Belgaum -590014, Karnataka.

LAB REPORT

On

ANALYSIS AND DESIGN OF ALGORITHMS (23CS4PCADA)

Submitted by

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in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING

B.M.S. COLLEGE OF ENGINEERING
(Autonomous Institution under VTU)
BENGALURU-560019
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CERTIFICATE

B. M. S. College of Engineering,
Bull Temple Road, Bangalore 560019
(Affiliated To Visvesvaraya Technological University, Belgaum)
Department of Computer Science and Engineering

This is to certify that the Lab work entitled **"ANALYSIS AND DESIGN OF ALGORITHMS"** carried out by Ludana Joshi **(1BF24CS156)**, who is bonafide student of **B. M. S. College of Engineering**. It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum during the year 2025-26. The Lab report has been approved as it satisfies the academic requirements in respect of Analysis and Design of Algorithms Lab - **(23CS4PCADA)** work prescribed for the said degree.

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Index Sheet

Course outcomes:

CO1	Analyze time complexity of algorithms using asymptotic notations.
CO2	Apply various algorithm design techniques for the given problem.
CO3	Evaluate the appropriate algorithm/design technique for solving a given problem.
CO4	Conduct practical experiments by applying appropriate algorithm design technique to solve problems.

Lab program 1:

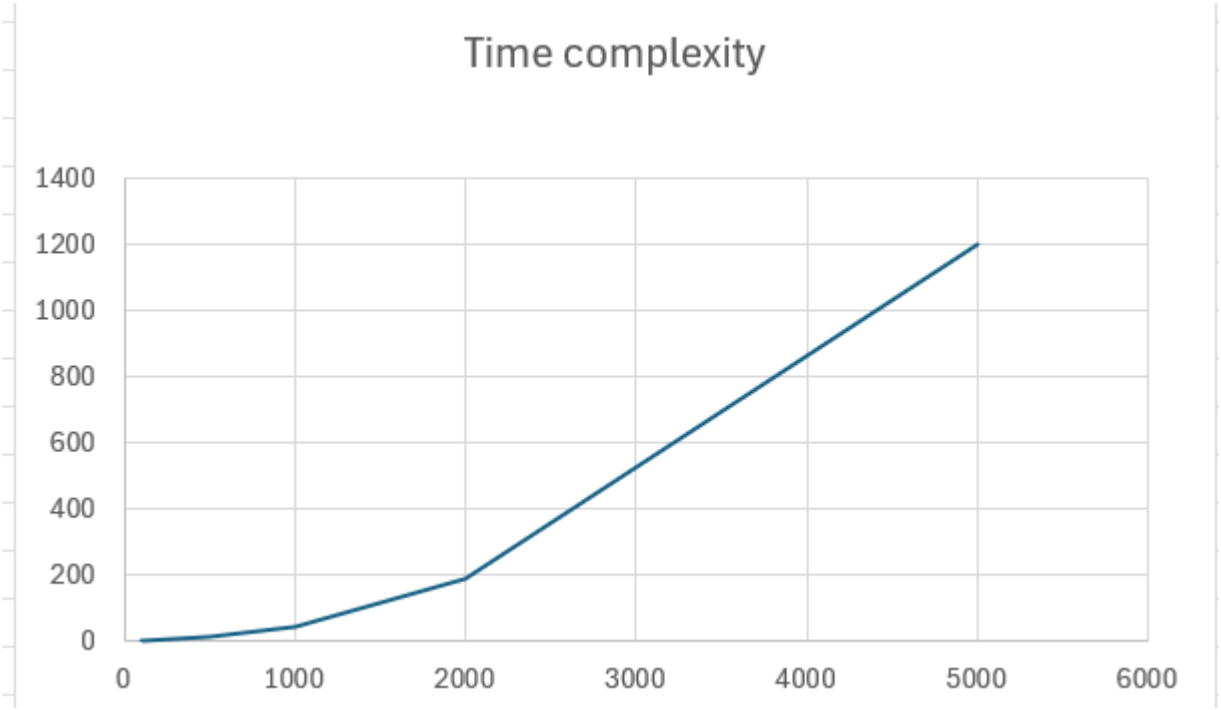
Write program to obtain the Topological ordering of vertices in a given digraph.

```
#include <stdio.h>

#define MAX 100

int graph[MAX][MAX];
int visited[MAX];
int stack[MAX];
int top = -1;

void dfs(int v, int n) {

   visited[v] = 1;

   for(int i = 0; i < n; i++) {

      if(graph[v][i] && !visited[i]) {
         dfs(i, n);
      }
   }

   stack[++top] = v;
}

void topologicalSort(int n) {

   for(int i = 0; i < n; i++) {

      if(!visited[i]) {
         dfs(i, n);
      }
   }

   printf("Topological Ordering:\n");

   while(top != -1) {
      printf("%d ", stack[top--]);
   }
}
```

```
int main() {

    int n, e;

    printf("Enter number of vertices: ");
    scanf("%d", &n);

    printf("Enter number of edges: ");
    scanf("%d", &e);

    printf("Enter edges (u v):\n");

    for(int i = 0; i < e; i++) {

        int u, v;

        scanf("%d %d", &u, &v);

        graph[u][v] = 1;
    }

    topologicalSort(n);

    return 0;
}
```

Output:

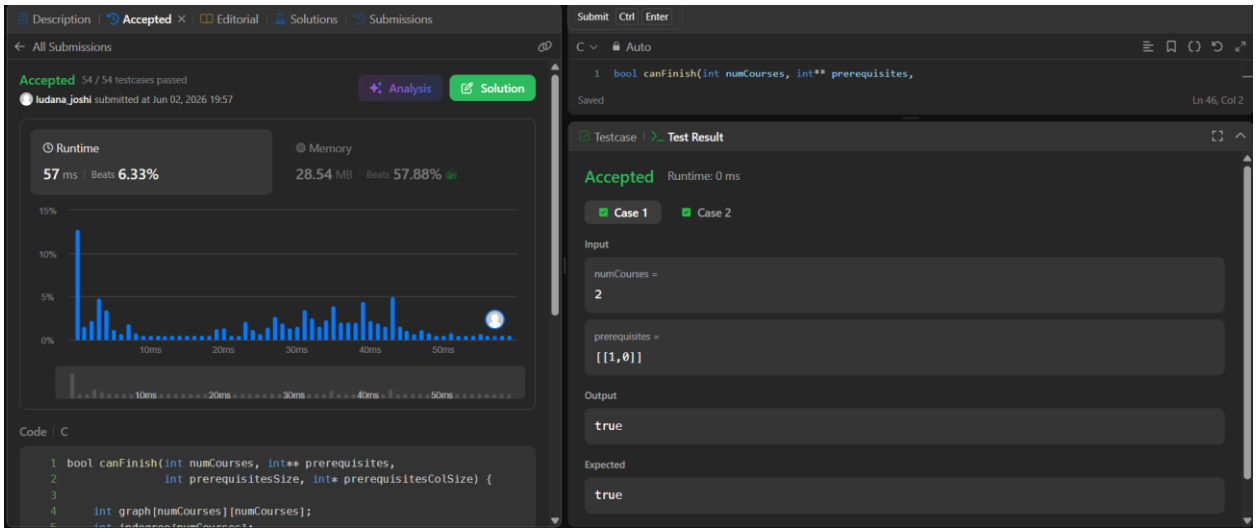
```
Enter number of vertices: 4
Enter number of edges: 4
Enter edges (u v):
0 1
0 2
1 3
2 3
Topological Ordering:
0 2 1 3
Process returned 0 (0x0)   execution time : 38.858 s
Press any key to continue.
```



Leetcode: 207. Course Schedule

```
bool canFinish(int numCourses, int** prerequisites,
            int prerequisitesSize, int* prerequisitesColSize) {

    int graph[numCourses][numCourses];
    int indegree[numCourses];

    for(int i=0;i<numCourses;i++) {
        indegree[i] = 0;
        for(int j=0;j<numCourses;j++)
            graph[i][j] = 0;
    }

    for(int i=0;i<prerequisitesSize;i++) {
        int a = prerequisites[i][0];
        int b = prerequisites[i][1];

        graph[b][a] = 1;
        indegree[a]++;
    }

    int queue[numCourses];
    int front = 0, rear = 0;

    for(int i=0;i<numCourses;i++) {
        if(indegree[i] == 0)
            queue[rear++] = i;
    }

    int count = 0;

    while(front < rear) {
        int course = queue[front++];
        count++;

        for(int i=0;i<numCourses;i++) {
            if(graph[course][i]) {
                indegree[i]--;

                if(indegree[i] == 0)
                    queue[rear++] = i;
            }
        }
    }

    return count == numCourses;
}
```

Output:


Description
Accepted
Editorial
Solutions
Submissions
All Submissions
Accepted 54 / 54 testcases passed
ludana_joshi submitted at Jun 02, 2026 19:57
Analysis
Solution
Runtime
57 ms | Beats 6.33%
Memory
28.54 MB | Beats 57.88%
15%
10%
5%
0%
10ms
20ms
30ms
40ms
50ms
Code | C
bool canFinish(int numCourses, int** prerequisites,
int prerequisitesSize, int* prerequisitesColSize) {
int graph[numCourses][numCourses];
Submit Ctrl Enter
C
Auto
bool canFinish(int numCourses, int** prerequisites,
Saved
Ln 46, Col 2
Testcase | Test Result
Accepted
Runtime: 0 ms
Case 1
Case 2
Input
numCourses =
2
prerequisites =
[[1,0]]
Output
true
Expected
true

Lab program 2:

Implement Johnson Trotter algorithm to generate permutations

```
#include <stdio.h>

#define LEFT 0
#define RIGHT 1

void print(int a[], int dir[], int n)
{
   for(int i = 0; i < n; i++)
   {
      if(dir[a[i]] == LEFT)
         printf("%d<- ", a[i]);
      else
         printf("%d-> ", a[i]);
   }
   printf("\n");
}

int getMobile(int a[], int dir[], int n)
{
   int mobile = 0;

   for(int i = 0; i < n; i++)
   {
      if(dir[a[i]] == LEFT && i > 0)
      {
         if(a[i] > a[i-1] && a[i] > mobile)
            mobile = a[i];
      }

      if(dir[a[i]] == RIGHT && i < n-1)
      {
         if(a[i] > a[i+1] && a[i] > mobile)
            mobile = a[i];
      }
   }
   return mobile;
}

int findPos(int a[], int n, int mobile)
```

```
{
   for(int i = 0; i < n; i++)
   {
      if(a[i] == mobile)
         return i;
   }
   return -1;
}

void johnsonTrotter(int n)
{
   int a[n], dir[n+1];

   for(int i = 0; i < n; i++)
   {
      a[i] = i + 1;
      dir[i + 1] = LEFT;
   }

   print(a, dir, n);

   while(1)
   {
      int mobile = getMobile(a, dir, n);

      if(mobile == 0)
         break;

      int pos = findPos(a, n, mobile);

      if(dir[mobile] == LEFT)
      {
         int temp = a[pos];
         a[pos] = a[pos - 1];
         a[pos - 1] = temp;
         pos = pos - 1;
      }
      else
      {
         int temp = a[pos];
         a[pos] = a[pos + 1];
         a[pos + 1] = temp;
```

```
            pos = pos + 1;
        }

        for(int i = 1; i <= n; i++)
        {
            if(i > mobile)
                dir[i] = !dir[i];
        }

        print(a, dir, n);
    }
}

int main()
{
    int n;

    printf("Enter n: ");
    scanf("%d", &n);

    johnsonTrotter(n);

    return 0;
}
```

Output:

```
Enter n: 3
1<- 2<- 3<-
1<- 3<- 2<-
3<- 1<- 2<-
3-> 2<- 1<-
2<- 3-> 1<-
2<- 1<- 3->

Process returned 0 (0x0)   execution time : 2.854 s
Press any key to continue.
```

Lab program 3:

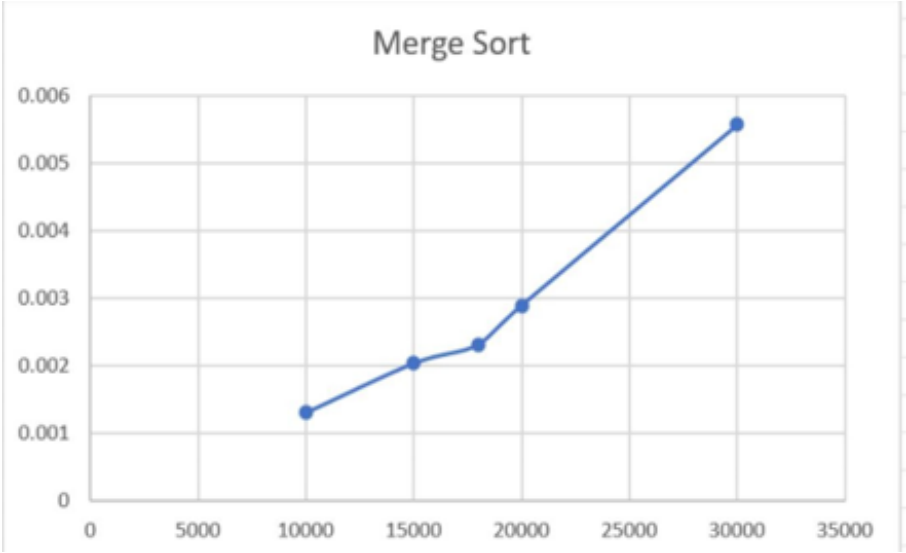
Sort a given set of N integer elements using Merge Sort technique and compute its time taken. Run the program for different values of N and record the time taken to sort.

```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>
void merge(int arr[], int lb, int mid, int ub)
{
   int n1 = mid - lb + 1;
   int n2 = ub - mid;

   int L[n1], R[n2];

   for (int i = 0; i < n1; i++)
      L[i] = arr[lb + i];

   for (int j = 0; j < n2; j++)
      R[j] = arr[mid + 1 + j];

   int i = 0, j = 0, k = lb;

   while (i < n1 && j < n2)
   {
      if (L[i] <= R[j])
         arr[k++] = L[i++];
      else
         arr[k++] = R[j++];
   }

   while (i < n1)
      arr[k++] = L[i++];

   while (j < n2)
      arr[k++] = R[j++];
}

void mergeSort(int arr[], int lb, int ub)
{
   if (lb < ub)
   {
      int mid = (lb + ub) / 2;
```

```
        mergeSort(arr, lb, mid);
        mergeSort(arr, mid + 1, ub);

        merge(arr, lb, mid, ub);
    }
}

int main()
{
    int n;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    int arr[n];

    printf("Enter elements:\n");

    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    clock_t start = clock();

    mergeSort(arr, 0, n - 1);

    clock_t end = clock();

    double time_taken = (double)(end - start) / CLOCKS_PER_SEC;

    printf("\nSorted array:\n");

    for (int i = 0; i < n; i++)
        printf("%d ", arr[i]);

    printf("\n");

    printf("Time taken: %.6f seconds\n", time_taken);

    return 0;
}
```

Output:

```
Enter number of elements: 5
Enter elements:
29 5 67 3 54

Sorted array:
3 5 29 54 67
Time taken: 0.000000 seconds

Process returned 0 (0x0)   execution time : 10.999 s
Press any key to continue.
```

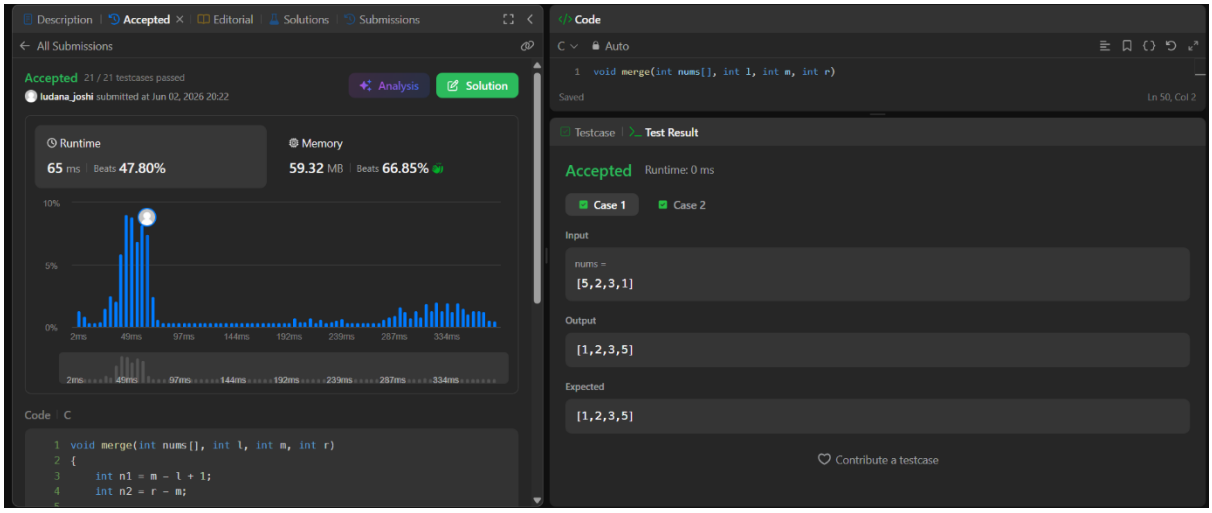


Leetcode 912: Sort an array

```
void merge(int nums[], int l, int m, int r)
{
    int n1 = m - l + 1;
    int n2 = r - m;

    int L[n1], R[n2];

    for(int i = 0; i < n1; i++)
        L[i] = nums[l + i];

    for(int j = 0; j < n2; j++)
        R[j] = nums[m + 1 + j];

    int i = 0, j = 0, k = l;

    while(i < n1 && j < n2)
    {
        if(L[i] <= R[j])
            nums[k++] = L[i++];
        else
            nums[k++] = R[j++];
    }

    while(i < n1)
        nums[k++] = L[i++];

    while(j < n2)
        nums[k++] = R[j++];
}

void mergeSort(int nums[], int l, int r)
{
    if(l < r)
    {
        int m = l + (r - l) / 2;

        mergeSort(nums, l, m);
        mergeSort(nums, m + 1, r);
```

```
        merge(nums, l, m, r);
    }
}

int* sortArray(int* nums, int numsSize, int* returnSize)
{
    mergeSort(nums, 0, numsSize - 1);

    *returnSize = numsSize;
    return nums;
}
```

Output:



Lab program 4:

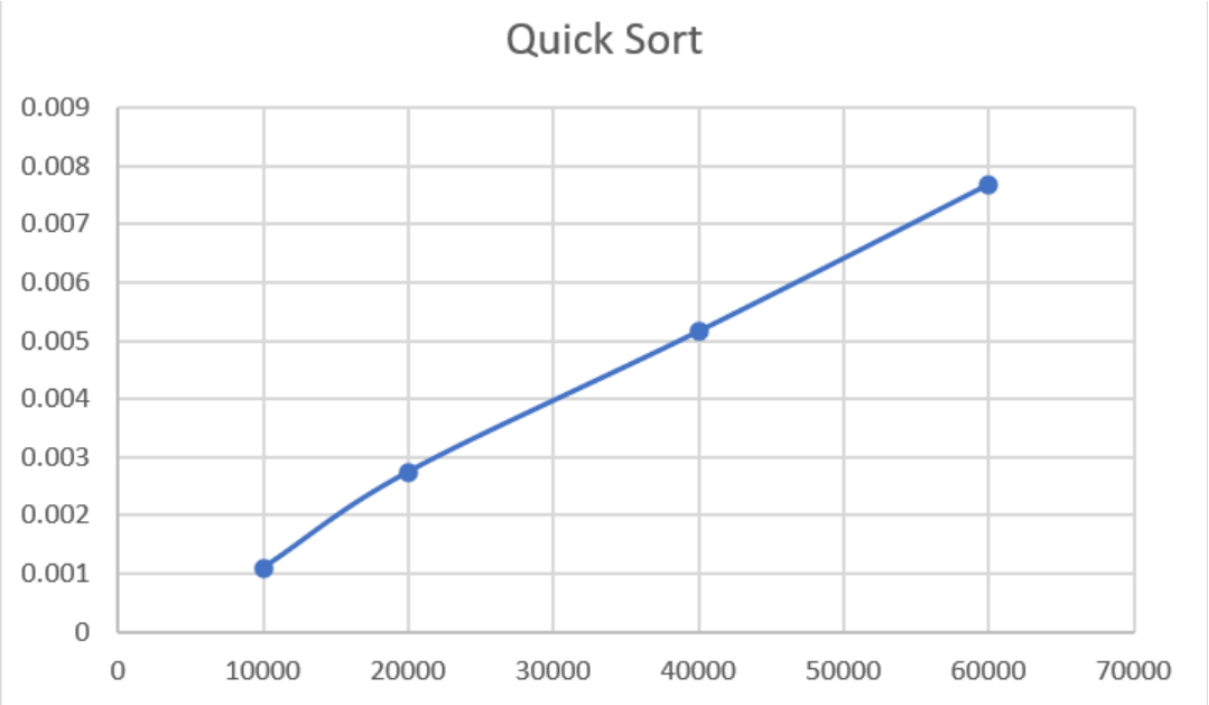
Sort a given set of N integer elements using Quick Sort technique and compute its time taken.

```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>

void swap(int *a, int *b)
{
   int temp = *a;
   *a = *b;
   *b = temp;
}

int partition(int arr[], int low, int high)
{
   int pivot = arr[high];
   int i = (low - 1);

   for (int j = low; j < high; j++)
   {
      if (arr[j] <= pivot)
      {
         i++;
         swap(&arr[i], &arr[j]);
      }
   }

   swap(&arr[i + 1], &arr[high]);
   return (i + 1);
}

void quickSort(int arr[], int low, int high)
{
   if (low < high)
   {
      int pi = partition(arr, low, high);

      quickSort(arr, low, pi - 1);
      quickSort(arr, pi + 1, high);
   }
}
```

```
int main()
{
   int N;

   printf("Enter number of elements: ");
   scanf("%d", &N);

   int arr[N];

   printf("Enter the elements:\n");

   for (int i = 0; i < N; i++)
   {
      scanf("%d", &arr[i]);
   }

   clock_t start, end;

   start = clock();

   quickSort(arr, 0, N - 1);

   end = clock();

   printf("Sorted array:\n");

   for (int i = 0; i < N; i++)
   {
      printf("%d ", arr[i]);
   }

   double time_taken = ((double)(end - start)) / CLOCKS_PER_SEC;

   printf("\nTime taken: %f seconds\n", time_taken);

   return 0;
}
```

Output:

```
Enter number of elements: 8
Enter the elements:
10 3 7 2 9 1 5 8
Sorted array:
1 2 3 5 7 8 9 10
Time taken: 0.000000 seconds
```

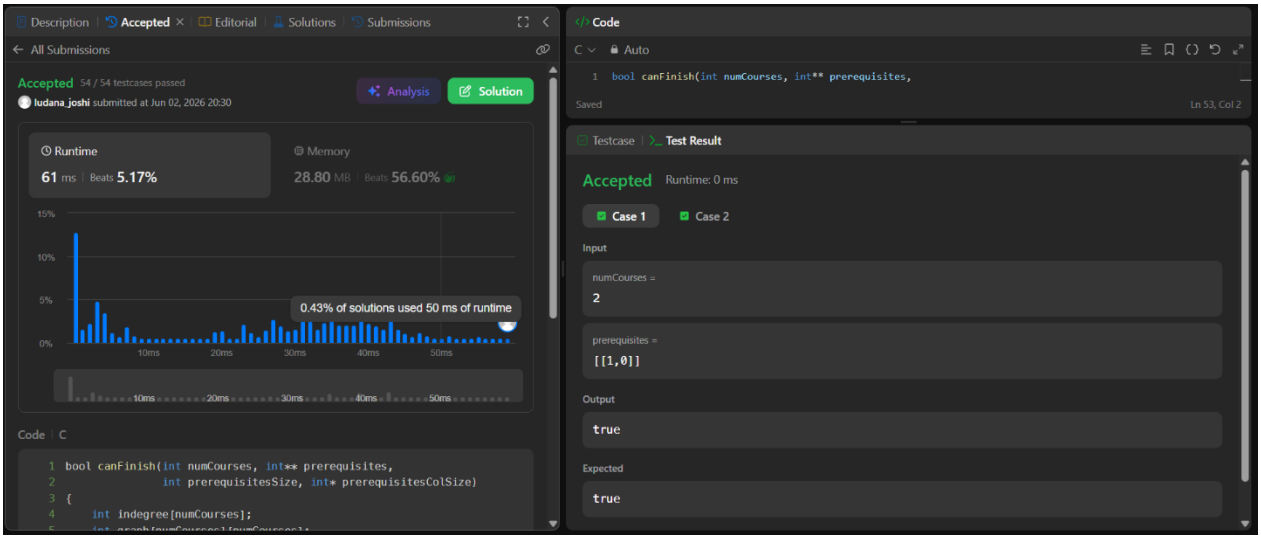


Leetcode 207: Course Schedule

```
bool canFinish(int numCourses, int** prerequisites,
            int prerequisitesSize, int* prerequisitesColSize)
{
   int indegree[numCourses];
   int graph[numCourses][numCourses];

   for(int i = 0; i < numCourses; i++)
   {
      indegree[i] = 0;

      for(int j = 0; j < numCourses; j++)
         graph[i][j] = 0;
   }

   for(int i = 0; i < prerequisitesSize; i++)
   {
      int a = prerequisites[i][0];
      int b = prerequisites[i][1];

      graph[b][a] = 1;
      indegree[a]++;
   }

   int queue[numCourses];
   int front = 0, rear = 0;

   for(int i = 0; i < numCourses; i++)
   {
      if(indegree[i] == 0)
         queue[rear++] = i;
   }

   int count = 0;

   while(front < rear)
   {
      int curr = queue[front++];
      count++;

      for(int i = 0; i < numCourses; i++)
      {
         if(graph[curr][i])
```

```
            {
                indegree[i]--;

                if(indegree[i] == 0)
                    queue[rear++] = i;
            }
        }
    }

    return count == numCourses;
}
```

Output:



Lab program 5:

Sort a given set of N integer elements using Heap Sort technique and compute its time taken.

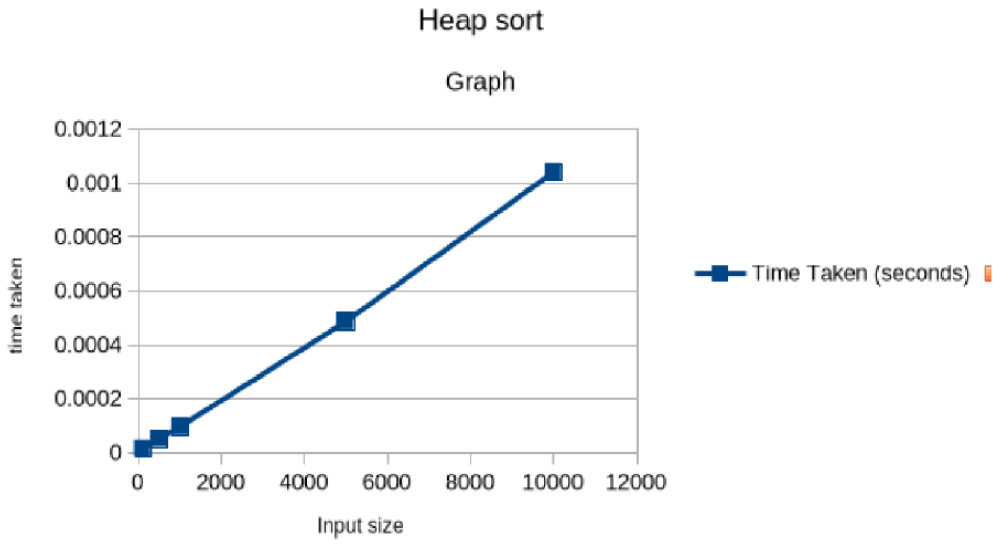
```
#include <stdio.h>

void swap(int *a, int *b)
{
   int temp = *a;
   *a = *b;
   *b = temp;
}

void siftUp(int a[], int i)
{
   while(i > 0)
   {
      int parent = (i - 1) / 2;

      if(a[parent] < a[i])
      {
         swap(&a[parent], &a[i]);
         i = parent;
      }
      else
      {
         break;
      }
   }
}

void buildHeap(int a[], int n)
{
   for(int i = 1; i < n; i++)
   {
      siftUp(a, i);
   }
}

void heapify(int a[], int n, int i)
{
   int largest = i;
   int left = 2 * i + 1;
```

```
    int right = 2 * i + 2;

    if(left < n && a[left] > a[largest])
        largest = left;

    if(right < n && a[right] > a[largest])
        largest = right;

    if(largest != i)
    {
        swap(&a[i], &a[largest]);
        heapify(a, n, largest);
    }
}

void heapSort(int a[], int n)
{
    buildHeap(a, n);

    for(int i = n - 1; i > 0; i--)
    {
        swap(&a[0], &a[i]);
        heapify(a, i, 0);
    }
}

void print(int a[], int n)
{
    for(int i = 0; i < n; i++)
        printf("%d ", a[i]);

    printf("\n");
}

int main()
{
    int n;

    printf("Enter n: ");
    scanf("%d", &n);

    int a[n];
```

```
    printf("Enter elements:\n");

    for(int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    heapSort(a, n);

    printf("Sorted array:\n");
    print(a, n);

    return 0;
}
```

Output:

```
Enter n: 6
Enter elements:
6
10
5
80
25
90
Sorted array:
5 6 10 25 80 90

Process returned 0 (0x0)   execution time : 49.457 s
Press any key to continue.
```



Lab program 6:

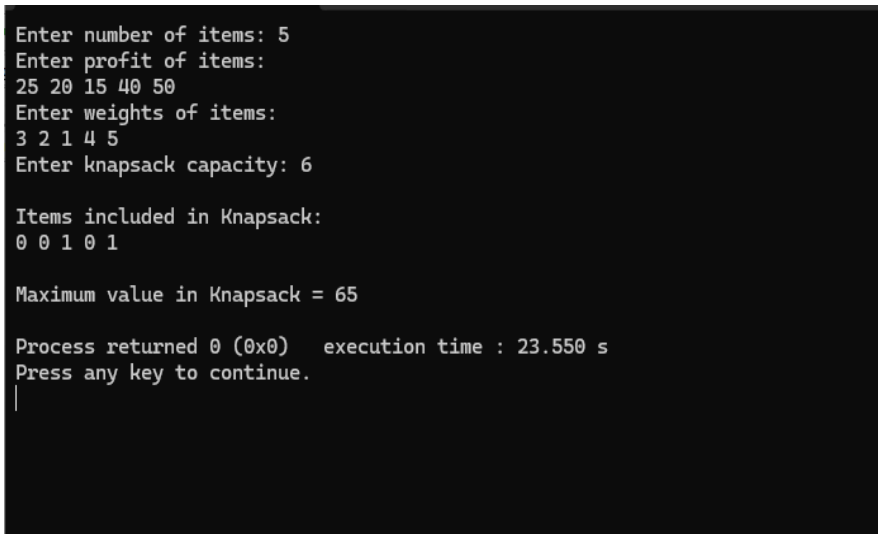
Implement 0/1 Knapsack problem using dynamic programming.

```
#include <stdio.h>

int knapsack(int W, int wt[], int val[], int n)
{
   int dp[n+1][W+1];

   for (int i = 0; i <= n; i++)
   {
      for (int w = 0; w <= W; w++)
      {
         if (i == 0 || w == 0)
         {
            dp[i][w] = 0;
         }
         else if (wt[i-1] <= w)
         {
            int include = val[i-1] + dp[i-1][w - wt[i-1]];
            int exclude = dp[i-1][w];

            dp[i][w] = (include > exclude) ? include : exclude;
         }
         else
         {
            dp[i][w] = dp[i-1][w];
         }
      }
   }

   int selected[n];

   for (int i = 0; i < n; i++)
      selected[i] = 0;

   int i = n;
   int w = W;

   while (i > 0 && w > 0)
   {
```

```
        if (dp[i][w] != dp[i-1][w])
        {
            selected[i-1] = 1;
            w = w - wt[i-1];
        }

        i--;
    }

    printf("\nItems included in Knapsack:\n");

    for (int i = 0; i < n; i++)
    {
        printf("%d ", selected[i]);
    }

    printf("\n");

    return dp[n][W];
}

int main()
{
    int n, W;

    printf("Enter number of items: ");
    scanf("%d", &n);

    int val[n], wt[n];

    printf("Enter profit of items:\n");
    for (int i = 0; i < n; i++)
    {
        scanf("%d", &val[i]);
    }

    printf("Enter weights of items:\n");
    for (int i = 0; i < n; i++)
    {
        scanf("%d", &wt[i]);
    }

    printf("Enter knapsack capacity: ");
```

```
    scanf("%d", &W);

    int result = knapsack(W, wt, val, n);

    printf("\nMaximum value in Knapsack = %d\n", result);

    return 0;
}
```

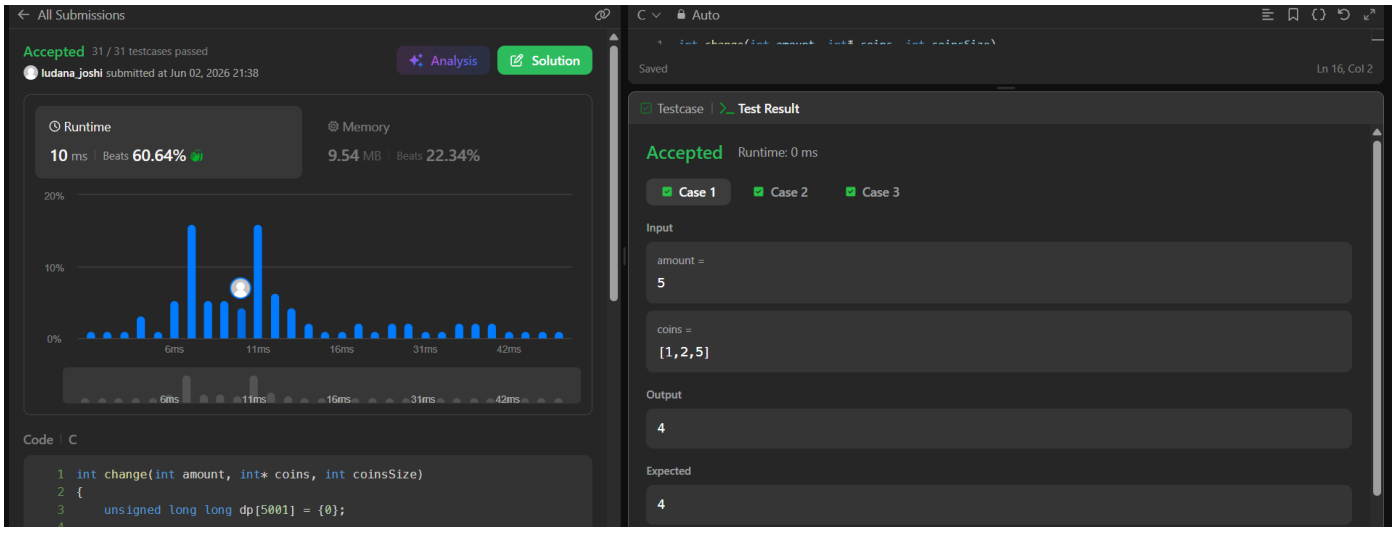
Output:



Leetcode : 518. Coin Change II

```
int change(int amount, int* coins, int coinsSize)
{
    unsigned long long dp[5001] = {0};

    dp[0] = 1;

    for(int i = 0; i < coinsSize; i++)
    {
        for(int j = coins[i]; j <= amount; j++)
        {
            dp[j] += dp[j - coins[i]];
        }
    }

    return (int)dp[amount];
}
```

Output:



Lab program 7:

Implement All Pair Shortest paths problem using Floyd's algorithm.

```
#include <stdio.h>

#define INF 99999
#define MAX 100

void printSolution(int dist[MAX][MAX], int V)
{
   printf("\nShortest distance matrix:\n");

   for (int i = 0; i < V; i++)
   {
      for (int j = 0; j < V; j++)
      {
         if (dist[i][j] == INF)
            printf("%7s", "INF");
         else
            printf("%7d", dist[i][j]);
      }
      printf("\n");
   }
}

void floydWarshall(int graph[MAX][MAX], int V)
{
   int dist[MAX][MAX];

   for (int i = 0; i < V; i++)
      for (int j = 0; j < V; j++)
         dist[i][j] = graph[i][j];

   for (int k = 0; k < V; k++)
   {
      for (int i = 0; i < V; i++)
      {
         for (int j = 0; j < V; j++)
         {
            if (dist[i][k] < INF &&
               dist[k][j] < INF &&
               dist[i][k] + dist[k][j] < dist[i][j])
            {
```

```
                dist[i][j] = dist[i][k] + dist[k][j];
            }
        }
    }
}

    for (int i = 0; i < V; i++)
    {
        if (dist[i][i] < 0)
        {
            printf("\nNegative cycle detected!\n");
            return;
        }
    }

    printSolution(dist, V);
}

int main()
{
    int V;
    int graph[MAX][MAX];

    printf("Enter number of vertices: ");
    scanf("%d", &V);

    printf("\nEnter adjacency matrix (-1 for no edge):\n");

    for (int i = 0; i < V; i++)
    {
        for (int j = 0; j < V; j++)
        {
            scanf("%d", &graph[i][j]);

            if (graph[i][j] == -1)
                graph[i][j] = INF;
        }
    }

    for (int i = 0; i < V; i++)
        graph[i][i] = 0;

    floydWarshall(graph, V);
```

```
    return 0;
}
```

Output:

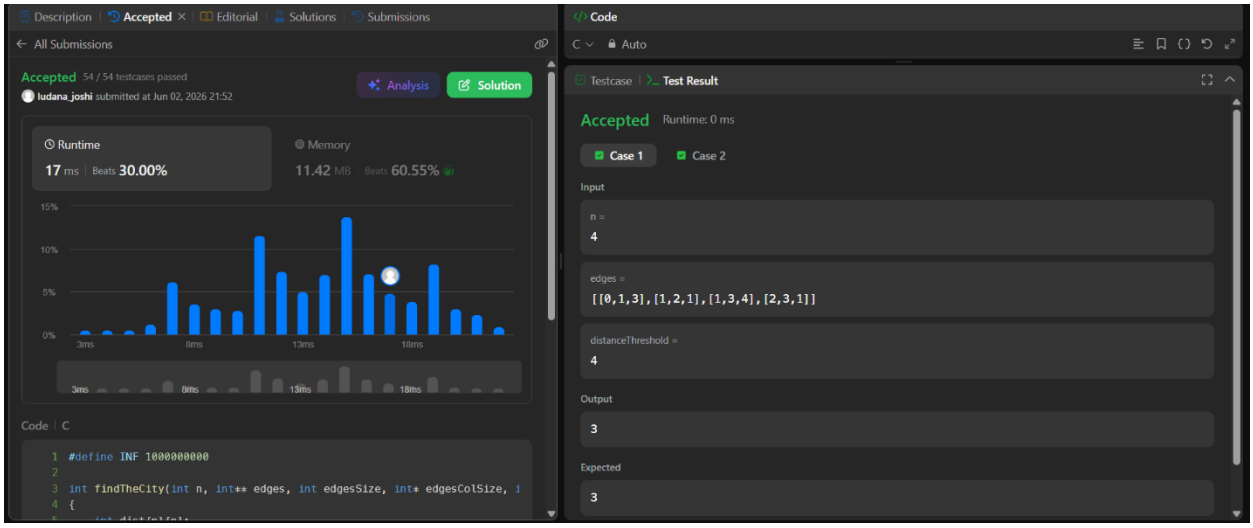
```
Enter number of vertices: 4

Enter adjacency matrix (-1 for no edge):
0 3 -1 5
2 0 -1 4
-1 6 0 -1
-1 -1 2 0

Shortest distance matrix:
      0      3      7      5
      2      0      6      4
      8      6      0     10
     10      8      2      0
```

Leetcode 1334: Find the city with the smallest no of neighbors at a threshold distance

```
#define INF 1000000000

int findTheCity(int n, int** edges, int edgesSize, int* edgesColSize, int distanceThreshold)
{
    int dist[n][n];

    for(int i = 0; i < n; i++)
    {
        for(int j = 0; j < n; j++)
        {
            if(i == j)
                dist[i][j] = 0;
            else
                dist[i][j] = INF;
        }
    }

    for(int i = 0; i < edgesSize; i++)
    {
        int u = edges[i][0];
        int v = edges[i][1];
        int wt = edges[i][2];

        dist[u][v] = wt;
        dist[v][u] = wt;
    }

    for(int k = 0; k < n; k++)
    {
        for(int i = 0; i < n; i++)
        {
            for(int j = 0; j < n; j++)
            {
                if(dist[i][k] != INF &&
                   dist[k][j] != INF &&
                   dist[i][k] + dist[k][j] < dist[i][j])
                {
                    dist[i][j] = dist[i][k] + dist[k][j];
                }
            }
        }
    }

    int minReachable = n;
    int answer = -1;

    for(int i = 0; i < n; i++)
    {
```

```
        int count = 0;

        for(int j = 0; j < n; j++)
        {
            if(i != j && dist[i][j] <= distanceThreshold)
                count++;
        }

        if(count <= minReachable)
        {
            minReachable = count;
            answer = i;
        }
    }

    return answer;
}
```

Output:



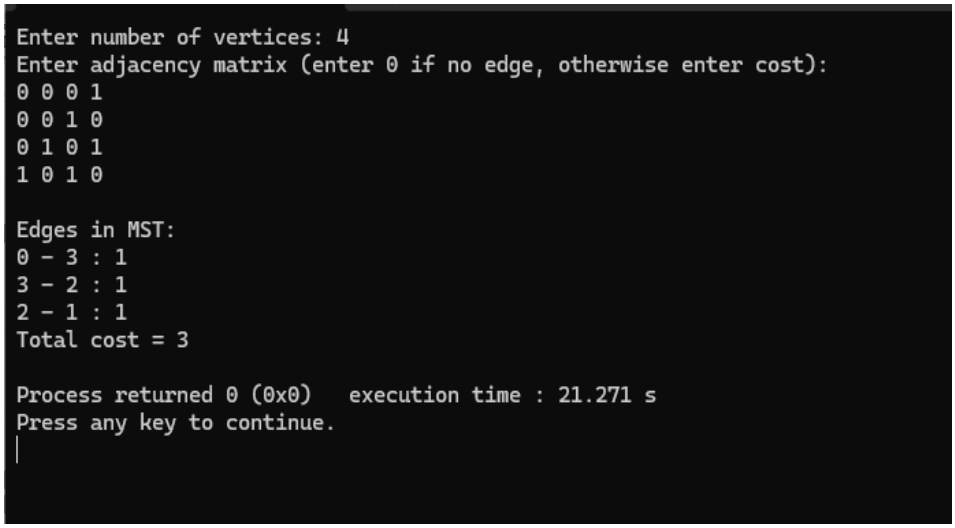
Lab program 8a:

Find Minimum Cost Spanning Tree of a given undirected graph using Prim's algorithm.

```
#include <stdio.h>
#include <limits.h>

int minKey(int key[], int mstSet[], int V)
{
   int min = INT_MAX, min_index = -1;

   for (int v = 0; v < V; v++)
      if (mstSet[v] == 0 && key[v] < min)
         min = key[v], min_index = v;

   return min_index;
}

void primMST(int graph[50][50], int V)
{
   int parent[V], key[V], mstSet[V];
   int totalCost = 0;

   for (int i = 0; i < V; i++)
   {
      key[i] = INT_MAX;
      mstSet[i] = 0;
   }

   key[0] = 0;
   parent[0] = -1;

   for (int count = 0; count < V - 1; count++)
   {
      int u = minKey(key, mstSet, V);
      mstSet[u] = 1;

      for (int v = 0; v < V; v++)
      {
         if (graph[u][v] && mstSet[v] == 0 && graph[u][v] < key[v])
         {
```

```
                parent[v] = u;
                key[v] = graph[u][v];
            }
        }
    }

    printf("\nEdges in MST:\n");

    for (int i = 1; i < V; i++)
    {
        printf("%d - %d : %d\n", parent[i], i, graph[i][parent[i]]);
        totalCost += graph[i][parent[i]];
    }

    printf("Total cost = %d\n", totalCost);
}

int main()
{
    int V;
    int graph[50][50];

    printf("Enter number of vertices: ");
    scanf("%d", &V);

    printf("Enter adjacency matrix (enter 0 if no edge, otherwise enter cost):\n");

    for (int i = 0; i < V; i++)
        for (int j = 0; j < V; j++)
            scanf("%d", &graph[i][j]);

    primMST(graph, V);

    return 0;
}
```

Output:


Enter number of vertices: 4
Enter adjacency matrix (enter 0 if no edge, otherwise enter cost):
0 0 0 1
0 0 1 0
0 1 0 1
1 0 1 0
Edges in MST:
0 - 3 : 1
3 - 2 : 1
2 - 1 : 1
Total cost = 3
Process returned 0 (0x0) execution time : 21.271 s
Press any key to continue.

Lab program 8b:

Find Minimum Cost Spanning Tree of a given undirected graph using Kruskal's algorithm.

```
#include <stdio.h>
#include <stdlib.h>

struct Edge
{
   int src, dest, weight;
};

struct Subset
{
   int parent;
};

int find(struct Subset subsets[], int i)
{
   while (subsets[i].parent != i)
      i = subsets[i].parent;

   return i;
}

void Union(struct Subset subsets[], int x, int y)
{
   int xroot = find(subsets, x);
   int yroot = find(subsets, y);

   subsets[xroot].parent = yroot;
}

int compare(const void *a, const void *b)
{
   return ((struct Edge *)a)->weight - ((struct Edge *)b)->weight;
}

void KruskalMST(struct Edge edges[], int V, int E)
{
   struct Edge result[V];
```

```
    struct Subset subsets[V];

    qsort(edges, E, sizeof(struct Edge), compare);

    for (int i = 0; i < V; i++)
        subsets[i].parent = i;

    int e = 0;
    int i = 0;

    while (e < V - 1 && i < E)
    {
        struct Edge next = edges[i++];

        int x = find(subsets, next.src);
        int y = find(subsets, next.dest);

        if (x != y)
        {
            result[e++] = next;
            Union(subsets, x, y);
        }
    }

    printf("\nEdges in MST:\n");

    int total = 0;

    for (i = 0; i < e; i++)
    {
        printf("%d -- %d == %d\n",
                result[i].src,
                result[i].dest,
                result[i].weight);

        total += result[i].weight;
    }

    printf("Total Cost = %d\n", total);
}

int main()
{
```

```
    int V;

    printf("Enter number of vertices: ");
    scanf("%d", &V);

    int E = V * (V - 1) / 2;

    struct Edge edges[E];

    printf("Enter edges (src dest weight):\n");

    for (int i = 0; i < E; i++)
    {
        scanf("%d %d %d",
              &edges[i].src,
              &edges[i].dest,
              &edges[i].weight);
    }

    KruskalMST(edges, V, E);

    return 0;
}
```

Output:

```
Enter number of vertices: 4
Enter edges (src dest weight):
0 1 10
0 2 6
0 3 5
1 2 15
1 3 4
2 3 2

Edges in MST:
2 -- 3 == 2
1 -- 3 == 4
0 -- 3 == 5
Total Cost = 11
```

Lab program 9:

Implement Fractional Knapsack using Greedy technique

```
#include <stdio.h>

struct Item
{
   int weight;
   int value;
};

int main()
{
   int n, capacity;

   printf("Enter number of items: ");
   scanf("%d", &n);

   struct Item items[n];

   for(int i = 0; i < n; i++)
   {
      printf("Enter weight and value of item %d: ", i + 1);
      scanf("%d %d", &items[i].weight, &items[i].value);
   }

   printf("Enter knapsack capacity: ");
   scanf("%d", &capacity);

   double ratio[n];

   for(int i = 0; i < n; i++)
      ratio[i] = (double)items[i].value / items[i].weight;

   for(int i = 0; i < n - 1; i++)
   {
      for(int j = i + 1; j < n; j++)
      {
         if(ratio[i] < ratio[j])
         {
            double tempR = ratio[i];
```

```
            ratio[i] = ratio[j];

            ratio[j] = tempR;

            struct Item temp = items[i];
            items[i] = items[j];
            items[j] = temp;
        }
    }
}

double maxValue = 0.0;
int remaining = capacity;

for(int i = 0; i < n; i++)
{
    if(items[i].weight <= remaining)
    {
        remaining -= items[i].weight;
        maxValue += items[i].value;
    }
    else
    {
        maxValue += ratio[i] * remaining;
        break;
    }
}

printf("Maximum value in Knapsack = %.2f\n", maxValue);

return 0;
}
```

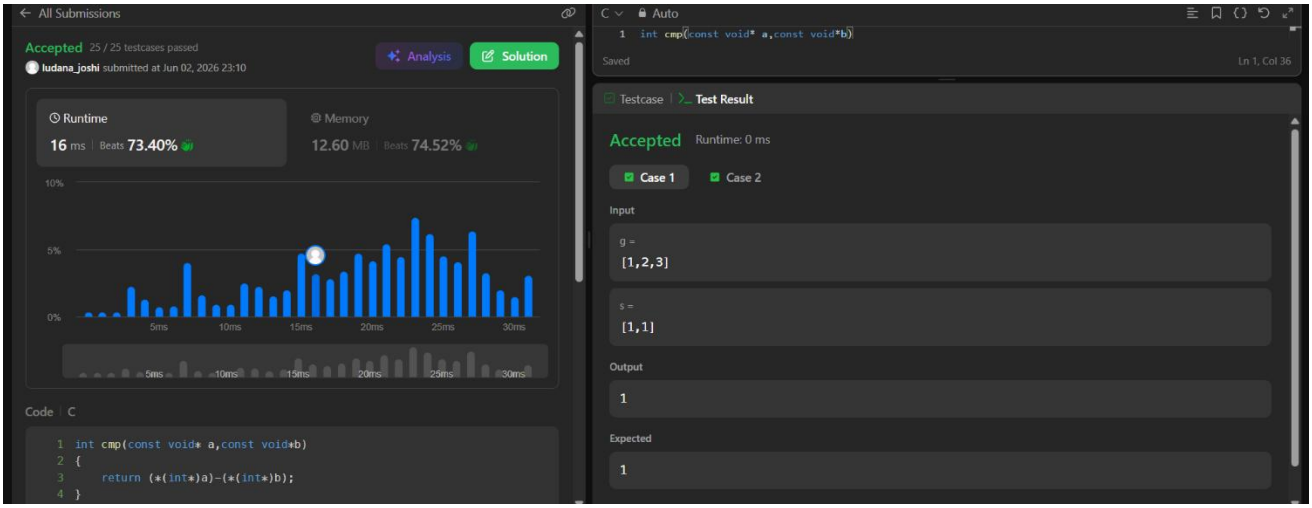
Output:

```
Enter number of items: 7
Enter weight and value of item 1: 1 5
Enter weight and value of item 2: 3 10
Enter weight and value of item 3: 5 15
Enter weight and value of item 4: 4 7
Enter weight and value of item 5: 1 8
Enter weight and value of item 6: 3 9
Enter weight and value of item 7: 2 4
Enter knapsack capacity: 15
Maximum value in Knapsack = 51.00
```

Leetcode 455:Assign Cookies

```
int cmp(const void* a,const void*b)
{
   return (*(int*)a)-(*(int*)b);
}
int findContentChildren(int* g, int gSize, int* s, int sSize) {
   int count=0;
   qsort(g,gSize,sizeof(int),cmp);
   qsort(s,sSize,sizeof(int),cmp);
   int i=0,j=0;
   while(i<gSize && j<sSize){
      if(g[i]<=s[j]){
         count++;
         i++;
      }
      j++;
   }
   return count;
}
```

Output:



Lab program 10:

From a given vertex in a weighted connected graph, find shortest paths to other vertices using Dijkstra's algorithm.

```
#include <stdio.h>
#include <limits.h>

#define V 10

int minDistance(int dist[], int sptSet[], int vertices)
{
   int min = INT_MAX, min_index;

   for (int v = 0; v < vertices; v++)
   {
      if (sptSet[v] == 0 && dist[v] <= min)
      {
         min = dist[v];
         min_index = v;
      }
   }

   return min_index;
}

void printSolution(int dist[], int vertices, int src)
{
   printf("Vertex \t Distance from Source %d\n", src);

   for (int i = 0; i < vertices; i++)
   {
      printf("%d \t %d\n", i, dist[i]);
   }
}

void dijkstra(int graph[V][V], int src, int vertices)
{
   int dist[vertices];
   int sptSet[vertices];

   for (int i = 0; i < vertices; i++)
   {
```

```
        dist[i] = INT_MAX;
        sptSet[i] = 0;
    }

    dist[src] = 0;

    for (int count = 0; count < vertices - 1; count++)
    {
        int u = minDistance(dist, sptSet, vertices);
        sptSet[u] = 1;

        for (int v = 0; v < vertices; v++)
        {
            if (!sptSet[v] &&
                graph[u][v] &&
                dist[u] != INT_MAX &&
                dist[u] + graph[u][v] < dist[v])
            {
                dist[v] = dist[u] + graph[u][v];
            }
        }
    }

    printSolution(dist, vertices, src);
}

int main()
{
    int vertices;

    printf("Enter number of vertices: ");
    scanf("%d", &vertices);

    int graph[V][V];

    printf("Enter adjacency matrix (0 if no edge):\n");

    for (int i = 0; i < vertices; i++)
    {
        for (int j = 0; j < vertices; j++)
        {
            scanf("%d", &graph[i][j]);
        }
```

```
    }

    int src;

    printf("Enter source vertex: ");
    scanf("%d", &src);

    dijkstra(graph, src, vertices);

    return 0;
}
```

Output:

```
Enter number of vertices: 5
Enter adjacency matrix (0 if no edge):
0 10 0 5 0
0 0 1 2 0
0 0 0 0 4
0 3 9 0 2
7 0 6 0 0
Enter source vertex: 0
Vertex   Distance from Source 0
0        0
1        8
2        9
3        5
4        7
```

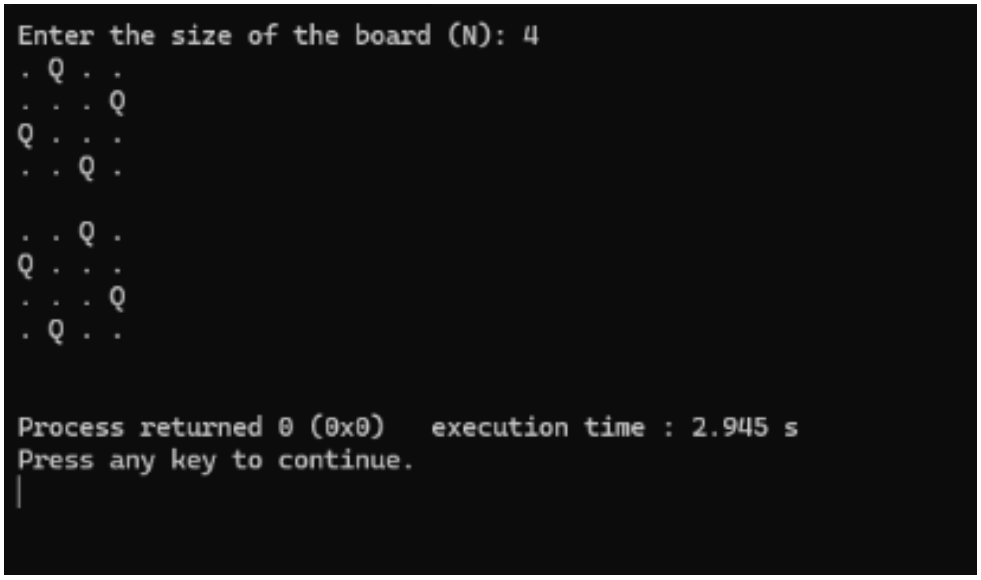
Lab program 11:

Implement "N-Queens Problem" using Backtracking.

```
#include <stdio.h>
#include <stdbool.h>

#define MAX 20

int N;
int board[MAX][MAX];

bool isSafe(int row, int col)
{
   for(int i = 0; i < row; i++)
   {
      if(board[i][col])
         return false;

      if(col - (row - i) >= 0 &&
        board[i][col - (row - i)])
         return false;

      if(col + (row - i) < N &&
        board[i][col + (row - i)])
         return false;
   }

   return true;
}

void printBoard()
{
   for(int i = 0; i < N; i++)
   {
      for(int j = 0; j < N; j++)
         printf("%c ", board[i][j] ? 'Q' : '.');

      printf("\n");
   }

   printf("\n");
```

```
}

void solve(int row)
{
   if(row == N)
   {
      printBoard();
      return;
   }

   for(int col = 0; col < N; col++)
   {
      if(isSafe(row, col))
      {
         board[row][col] = 1;

         solve(row + 1);

         board[row][col] = 0;
      }
   }
}

int main()
{
   printf("Enter the size of the board (N): ");
   scanf("%d", &N);

   if(N > MAX || N < 1)
   {
      printf("Invalid board size. Please enter between 1 and %d.\n", MAX);
      return 1;
   }

   solve(0);

   return 0;
}
```

Output:


Enter the size of the board (N): 4
. Q . .
. . . Q
Q . . .
. . Q .
. . Q .
Q . . .
. . . Q
. Q . .
Process returned 0 (0x0) execution time : 2.945 s
Press any key to continue.